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Question 01 [2 + 1 + 2 = 5 Marks]

You are given the polynomial equation:

$$f(x) = 0.5(x + 8)^2 + 3$$

- Use the Polynomial class of Numpy library to represent the given polynomial. Print the polynomial equation.
- Calculate and display the degree of the polynomial.
- Compute the value of the expression $f(x).f'(x)$ at $x=4$.

Question 02 [3 marks]

In class, we constructed $X=$

$$\begin{bmatrix} x_0^0 & x_0^1 & \dots & x_0^{n-1} & x_0^n \\ x_1^0 & x_1^1 & \dots & x_1^{n-1} & x_1^n \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n-1}^0 & x_{n-1}^1 & \dots & x_{n-1}^{n-1} & x_{n-1}^n \\ x_n^0 & x_n^1 & \dots & x_n^{n-1} & x_n^n \end{bmatrix}$$

Now, we have to build a scaled geometric matrix. Given $n+1$ data points define the X_{geo} matrix as:

$$X_{geo} = x_i^{2j} \quad i, j = 0, \dots, n$$

$$\begin{bmatrix} x_0^0 & x_0^2 & \dots & x_0^{2n-2} & x_0^{2n} \\ x_1^0 & x_1^2 & \dots & x_1^{2n-2} & x_1^{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n-1}^0 & x_{n-1}^2 & \dots & x_{n-1}^{2n-2} & x_{n-1}^{2n} \\ x_n^0 & x_n^2 & \dots & x_n^{2n-2} & x_n^{2n} \end{bmatrix}$$

Write a **function** that takes **data_x** as a parameter and populates the X_{new} matrix using (0,5), (1,7), (2,9), (3,11) data points. Your output has to show the newly implemented Vandermonde matrix clearly.

Question 03 [2 + 2 = 4 Marks]

You are given the following table,

X values	Corresponding $f(x)$
1	2
3	12
5	30

You already know how to find out the interpolating polynomial using the **Lagrange Interpolation Method**.

Now, instead of the conventional way of Lagrange Polynomial Interpolation, you will use a new form of it.

General form of the new n-degree Lagrange polynomial:

$$p_n(x) = \sum_{k=0}^n f(x_k) l_k(x)$$

where

$$l_k(x) = \prod_{j=0, j \neq k}^n \frac{x_k - x_j}{x_k - x}$$

a) Evaluate the Lagrange bases $[l_0(x), l_1(x), l_2(x)]$ at $x=7$ for a polynomial that passes through the given data points.

b) Use the basis polynomials to calculate the interpolated value $P(7)$ using the Lagrange Interpolating Polynomial.

Question 04 [2 + 1 = 3 Marks]

Suppose there is a function,

$$g(x) = 2x^3 - 5x^2 + 4x + 8$$

You have to investigate the behavior of the **Forward Difference Method** for approximating $g'(x)$ at the point $x = 2$. Specifically, determine the **smallest step size h** such that the **absolute error remains below a tolerance of 0.0015**. Your output should clearly state the **minimum value of h** and the **absolute error**.

As you already know,

- Central Difference, $g'(x) = (g(x+h) - g(x))/h$
- Absolute error = |Actual derivative - Approximate derivative|

You should start with an **initial step size of $h = 0.5$** and iteratively **divide the value of h by 4** until the error condition is satisfied.

Hint: You have to use the polynomial class from the numpy library to define the function, $f(x)$.